MATRIC MASTERY AI

v4.0.0 — AI Workspace Upgrade Brief

Independent scrolling • Student guidance • Subject information • Free-tier transparency

Builder: Gemini AI Studio
Content & product direction: ChatGPT Project Content Leader
Project: Matric Mastery PK

Purpose
Upgrade the existing AI workspace into a complete, student-friendly tutoring destination while preserving the working Cloudflare Worker integration and all existing chat capabilities.

IMPLEMENTATION INSTRUCTIONS

This document is the consolidated implementation brief. Follow the requirements in order, prioritize the independent scrolling fix, and do not deploy automatically.

1. Project objective

You are the Builder for Matric Mastery PK. This is a focused improvement to the existing v3.9.0 AI workspace, with the next major release targeted as **v4.0.0**. Do not rebuild the entire website.

The owner has identified two requirements:

1. **Independent scrolling:** the AI conversation and surrounding AI page must scroll separately.
2. **More useful information:** the page must explain what the tutor offers, how students should use it, and what the free-tier limitations are.

Preserve the existing premium black-and-yellow Matric Mastery identity.

2. Critical: fix independent scrolling

This is the highest priority. Currently, scrolling the AI conversation also moves the surrounding webpage. Implement two genuinely independent scrolling regions.

A. Main AI page

- The overall AI page must have its own clearly defined layout and scrolling behaviour.
- Header, information sections, usage guide, and other page content must remain separate from the conversation scroll container.
- Scrolling the information page must not move the conversation internally.
- Avoid unnecessary nested scrolling outside the actual chat workspace.

B. AI conversation

- Give the conversation transcript its own dedicated scroll container.
- Only the transcript should scroll when the user swipes or scrolls inside the conversation.
- Keep the header and composer in their intended positions.
- Automatically scroll to the latest message only when appropriate.
- Do not forcibly jump to the bottom when a student is reading older messages.
- Preserve the user's scroll position while they inspect older messages.
- Use appropriate overscroll containment so reaching the transcript boundary does not unintentionally scroll the outer page.

C. Mobile behaviour

- Test especially carefully on Android.
- Swiping inside chat must move only the chat transcript.
- Swiping outside chat must move only the information page.
- Keep the composer reachable and prevent keyboard overlap or clipped content.
- Do not solve the problem by disabling scrolling everywhere. The goal is predictable, independent scrolling.

3. Upgrade the AI page into a complete tutoring destination

The AI page must explain the service before and during use. Add a well-organized information experience around the existing chat workspace. Avoid a long wall of text; use refined sections, compact cards, helpful descriptions, and clear visual hierarchy.

A. Welcome and introduction

Introduce Matric Mastery AI as: **“Your personal AI study assistant for Punjab Board Class 9–10.”**

- Understand difficult concepts.
- Solve mathematical problems step by step.
- Learn Physics formulas and numerical problems.
- Understand Chemistry reactions and concepts.
- Study Biology definitions and processes.
- Ask follow-up questions when an explanation is unclear.

Make clear that the AI is a learning assistant, not an official Punjab Board service.

B. Subjects

Mathematics

- Algebra and equations
- Factorization
- Geometry
- Trigonometry
- Step-by-step numerical solutions

Physics

- Concepts and definitions
- Formulas and units
- Numericals
- Laws and applications

Chemistry

- Chemical equations
- Balancing reactions
- Atomic structure
- Chemical bonding
- Concepts and definitions

Biology

- Cell structure
- Human body systems
- Photosynthesis
- Definitions and biological processes

Do not imply that every topic has been independently verified against every current board textbook.

C. How to use Matric Mastery AI

1. Select your preferred language: English or Urdu.
2. Select the relevant subject when applicable.
3. Write your question clearly.
4. For numericals, include all given values and units.
5. Ask for a step-by-step explanation if needed.

6. Read the solution and ask follow-up questions.
7. If something is unclear, ask the AI to explain it more simply.

Useful example prompts

- “Explain this topic in simple words.”
- “Solve this equation step by step.”
- “Explain this Physics numerical with formulas.”
- “Balance this chemical equation.”
- “Explain this in simple Pakistani Urdu.”
- “Give me a short exam-style answer.”

Make examples interactive where practical.

D. Tips for better answers

- Ask one clear question at a time.
- Mention your class and subject.
- Include units in numerical problems.
- Specify whether you want a short answer or detailed explanation.
- Ask follow-up questions rather than repeating the same question.
- Verify important textbook definitions and exam requirements against prescribed material.

Student reminder: AI can occasionally make mistakes. Always verify important answers with your textbook and teacher.

4. Free-tier limit information

Add a clearly visible section titled **“Free AI Access — Usage Information.”**

- Matric Mastery AI currently uses a free-tier AI service.
- Usage is subject to provider limits and shared daily capacity.
- The AI may occasionally respond slowly.
- During heavy usage, students may temporarily encounter a usage-limit message.
- Availability is not guaranteed at every moment.
- A temporary limit does not mean the website or student's account is broken.
- Students should wait and try again later.

Accuracy requirements

Do not invent a guaranteed number of questions per day, a guaranteed number of tokens per student, a specific reset time, unlimited access, or guaranteed response speed.

Only display exact quota numbers or reset information if the backend provides authoritative, current data.

Live status display

Use the existing `/api/ai/health` information where appropriate. Distinguish between backend available, free-tier service operational, temporary usage limit reached, and service temporarily unavailable.

Do not label a healthy backend as proof that the user has unlimited remaining quota.

Friendly Urdu reminder

Matric Mastery AI ■■■ ■■■■■■ ■■ ■■■■ ■■ ■■■■■■■ ■■■■ ■■■■■ ■■ ■■ ■■■■■■■■■ ■■ ■■■■■■■ ■■ ■■■■■■ ■■■■ ■■■■■■ ■■■■■■■■■ ■■ ■■■■■■ ■■■■■■ ■■■■ ■■■■■■ ■■■■ ■■ ■■■■■■■ ■■■■■■■ ■■ ■■ ■■■■■ ■■ ■■■■■ ■■■■ ■■■■■ ■■■■■ ■■■■ ■■■■ ■■■■ ■■■■ ■■■■■■■ ■■■■■ ■■■■■■

Keep the reminder concise and student-friendly.

5. Chat workspace presentation

Keep the actual conversation experience clean. Information sections must not overwhelm the chat. Clearly separate the introduction and usage information, the actual AI tutoring workspace, and free-tier usage guidance.

Consider compact navigation or clearly labelled sections so students can quickly reach the chat. Do not make students scroll through a large introduction every time they want to ask a question.

Preserve all existing capabilities:

- English / Urdu selector and subject selection.
- Streaming responses and Stop generation.
- Copy, Share, Regenerate, and New Chat.
- Existing local history and JSON export.
- Error and retry handling.

Do not introduce fake functionality.

6. Design quality

- Near-black background and yellow `#FFD60A` accent.
- Clean typography, refined borders, excellent readability, and consistent spacing.
- Subtle animations and responsive layouts.
- Avoid excessive cards, repeated information, oversized decorative banners, unnecessary gradients, large empty sections, excessive rounded pills, and a cluttered dashboard appearance.

Information should feel like part of a premium educational product, not documentation pasted underneath a chatbot.

7. Technical boundaries

Preserve the current production architecture.

Worker: https://matric-mastery-ai-worker.boyc60844.workers.dev

Frontend: https://matric-mastery-pk.vercel.app

Continue using the existing `VITE_AI_WORKER_URL` resolver and Cloudflare Worker integration.

- Do not replace the AI model or introduce direct Gemini API requests.
- Do not modify Worker inference unnecessarily.
- Do not add paid services.
- Do not enable unapproved vision functionality.
- Do not change unrelated website pages or remove working chat functionality.
- Keep changes focused on the AI page and supporting components that genuinely need modification.

8. Required testing

Scrolling

- Desktop: scroll chat independently.
- Desktop: scroll information independently.
- Android: swipe inside chat without moving the outer page.
- Android: swipe outside chat without moving the transcript.
- Verify keyboard and composer behaviour.
- Verify no scroll trapping or horizontal overflow.

Information

- All four subjects are represented correctly.
- Instructions are understandable to Class 9–10 students.
- English and Urdu information are readable.
- Free-tier explanations contain no invented quota figures.

AI functionality

- Submit a Mathematics question.
- Submit a Physics question.
- Submit an Urdu question.
- Verify streaming and follow-up questions.
- Verify existing error and retry behaviour.

Regression

- Run production build and TypeScript checks.
- Run existing automated tests.
- Confirm the rest of the website remains unaffected.

9. Release management

After actual implementation and verification, update the release documentation and changelog for **v4.0.0**. Update the package version, release history, Website Guide, and relevant feature documentation consistently. Preserve historical entries.

Do not claim features that were not implemented or tested. Do not deploy automatically.

10. Final acceptance criteria

The result should feel like a complete AI learning destination. Students should be able to:

1. Understand what Matric Mastery AI offers.
2. Learn how to ask better questions.
3. Understand the free-tier limitations.
4. Open the tutor and use it without unnecessary navigation.
5. Scroll the conversation independently from the surrounding page.
6. Use everything comfortably on a small Android screen.

Prioritize the independent scrolling fix first. Do not sacrifice existing functionality to achieve the new design.